MSTAR PRACTICE WORKSHEET

Lesson 2-10: Choose Appropriate Measures

Grade 6 Mathematics · Standard 6.DS.6d

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

A gymnast's scores are: 8.5, 8.8, 8.7, 8.6, 8.9. There are no outliers. Which measure best represents a typical score?

- (A) Median
- (B) Neither — the data is too small
- (C) Both are equally bad
- (D) Mean

The data is symmetric with no outliers, so the mean (8.7) best represents the typical score.

Question 2 SELECTED RESPONSE — CORRECT: B

A runner's mile times are: 7:10, 7:15, 7:12, 7:20, 12:00. The 12:00 was due to a cramp. Which measure best represents a typical mile?

- (A) Mean — it uses all the data
- (B) Median — the outlier 12:00 pulls the mean too high
- (C) Mode — it shows the most common time
- (D) Neither measure works

The 12:00 is an outlier that pulls the mean up. The median (7:15) better represents the runner's typical time.

Question 3 SELECTED RESPONSE — CORRECT: C

Data set: 5, 6, 7, 7, 8, 50. The mean is about 13.8 and the median is 7. Which better represents the typical value?

- (A) Mean (13.8) — it includes all values
- (B) Both are equally good
- (C) Median (7) — the outlier 50 inflates the mean
- (D) Neither works for this data

> The mean (13.8) is higher than 5 of the 6 values because the outlier 50 pulls it up. The median (7) better represents a typical value.

Question 4 SELECTED RESPONSE — CORRECT: A

A baseball player's batting averages over 6 seasons are: .280, .295, .290, .285, .300, .110. The .110 was an injury-shortened season. Which measure better represents the player's typical batting average?

- (A) Median, because the .110 outlier pulls the mean down
- (B) Mean, because it uses all the data
- (C) Mean, because .110 is a real season
- (D) Neither measure works

> The .110 is an outlier that pulls the mean down to .260. The median (.2875) better represents typical performance because it isn't affected by the extreme value.

Question 5 SELECTED RESPONSE — CORRECT: C

A scout records points per game: 14, 16, 15, 17, 16, 58. To report a TYPICAL game, which measure of center is best?

- (A) Mean, because it uses every game
- (B) Range, because it shows the highest game
- (C) Median, because 58 is an outlier that pulls the mean up
- (D) Mode, because 16 appears twice

> The 58-point game is an outlier. It pulls the mean above 5 of the 6 games, so the median (16) better shows a typical game.

Question 6 SELECTED RESPONSE — CORRECT: C

The equipment manager needs to order the MOST COMMON jersey size from this list: M, L, M, S, M, L, M. Which measure answers the question?

- (A) Mean, because it averages the sizes
- (B) Median, because it finds the middle size
- (C) Mode, because it names the value that appears most
- (D) Range, because it shows the size gap

> The mode is the value that shows up most often. Here size M appears 4 times, so the mode (M) tells the manager what to order most.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A data set of house prices contains one mansion worth far more than every other home. Which pair of measures best describes this data?

- **A** Median and IQR
- **B** Mean and MAD
- **C** Mean and range
- **D** Mode and range

> The mansion is an outlier that pulls the mean upward and inflates the range. The median and IQR describe the middle of the data and are resistant to a single extreme value.

- **B:** The mansion drags the mean upward, and MAD measures distance from that pulled mean, so both are distorted.
- **C:** Range is the maximum minus the minimum, so the mansion sets it by itself, and it pulls the mean too.
- **D:** Mode is the most frequent price, which says nothing about the middle, and the range is set by the mansion alone.

Part B — correct answer: D

When are the mean and MAD the better choice?

- **A** Whenever the data set is large.
- **B** Whenever the values are whole numbers.
- **C** Only when the median cannot be found.
- **D** When the data is roughly symmetric with no outliers.

> With symmetric data and no extreme values, the mean sits at the center and uses every value, so the mean and MAD summarize the set well.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that correctly guide the choice of measures:

- **A** Skewed data with outliers is better summarized by the median and IQR.
- **B** Symmetric data with no outliers is well summarized by the mean and MAD.
- **C** The mean is resistant to outliers.
- **D** An outlier affects the range more than it affects the IQR.
- **E** The shape of the distribution should influence which measures you report.

> A, B, D, and E are true. C is false — the mean is strongly affected by outliers, which is precisely why the median is preferred for skewed data.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A local newspaper reports that the average home price in a neighborhood is $450,000. The actual prices are: $180K, $200K, $190K, $210K, $195K, $1,725K. Is the reporter's claim misleading? Calculate the mean and median and explain which better represents a 'typical' home price.

Model answer: The six prices add to $2,700K, so the mean is 2,700 ÷ 6 = $450K, matching the reporter — but the median is the average of the middle two prices, $195K and $200K, which is $197.5K. Five of the six homes cost between $180K and $210K, so the $1,725K home is an outlier pulling the mean far above what most homes cost. The claim is misleading, and the median better represents a typical price.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.